

SAMARTH PRATAP SINGH

AI/ML Engineer | LangGraph · Agentic Pipelines · RAG | 2027

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EDUCATION

VIT Bhopal University, Bhopal

2023 – 2027

B.Tech - Computer Science and Engineering - **CGPA - 8.57**

Bhopal, Madhya Pradesh

EXPERIENCE

Artificial Intelligence Applications Development Intern

May 2026 – Present

AmberFlux EdgeAI Private Limited | Remote

Hyderabad, Telangana

- Architected a vision extraction pipeline for multi-page architectural PDFs using GPT-5 — reduced latency from ~7 min to ~90 sec on 20-page documents via concurrent batch dispatch (`ThreadPoolExecutor` + `asyncio`), with structured output enforcement and retry handling.
- Built a cover and dimension intake pipeline combining non-AI heuristic extraction (regex/Docling) with conditional GPT-5 fallback — vision invoked only for missing fields, achieving confidence >0.85 and near-complete field extraction on real architectural lead sheets via token-efficient prompting.
- Designed a job aggregation layer consolidating multi-source document outputs into a structured JSON payload for documents up to 400 pages — enforcing page-level validation guardrails, drawing-page exclusion, and LangGraph-based routing for decision-driven pipeline execution.

PROJECTS

DoCopilot (RAG Document Q&A) 🗄️ | [Next.js](#), [FastAPI](#), [Qdrant](#) (Hybrid), [Reranking](#)

Dec 2025

- Built and deployed a full-stack RAG system — PDF/TXT ingestion, Qdrant hybrid search (BM25 + dense vectors) with RRF fusion, cross-encoder reranking via Cohere's hosted Rerank API (rerank-english-v3.0), and cited answer retrieval via FastAPI backend and Next.js frontend.
- Added multi-tenant document isolation via Qdrant payload-indexed filtering, migrated vector storage from in-memory to a persistent Qdrant service (Docker), and built a GitHub Actions regression suite gating every push against an 8%-tolerance drop from the 89.2% correctness baseline.
- Ran a 40-query LLM-as-Judge ablation study reporting **89.2%** avg correctness and **90.5%** relevance — a **31-point** improvement over a keyword-matching baseline (57.7%) — with **100%** source rate and **2.86s** avg latency.

Argus - Multi-Agent Research Engine 🤖 🗄️ | [FastAPI](#), [LangGraph](#), [Docker](#), [LangSmith](#)

Feb 2026

- Built a LangGraph supervisor-based multi-agent pipeline with 5 specialist agents (planner, researcher, critic, writer, supervisor) using LLM-driven `Command(goto)` routing to synthesize cited markdown reports across **3** depth tiers (**20s/45s/90s**).
- Designed an async submit → stream → fetch architecture using FastAPI Server-Sent Events (SSE) for real-time UI logging, backed by a dual-dialect (SQLite/Postgres) adapter with connection pooling and dynamic LangGraph checkpointing.
- Implemented a Human-in-the-Loop (HIL) review gate using LangGraph interrupts, capped at **3** critique iterations with a **30-min** auto-finalize timeout via background daemon thread, and rate-limited to **5 req/hour** to protect free-tier API quotas.

ContextCore (Stateful Memory Agent) 🗄️ | [LangGraph](#), [PostgreSQL](#), [Qdrant](#), [MongoDB](#), [MCP](#) Jun 2026

- Built a FastMCP server exposing task/note management as callable tools (create/list/update/delete, save/search), bound via a dedicated MCP client in the LangGraph agent executor.
- Designed a dual-memory architecture (PostgreSQL checkpointing, Qdrant semantic recall, MongoDB profiles) with an intent router dispatching queries via conditional LangGraph edges.
- Built an LLM-as-Judge eval harness reaching **100%** pass rate across 17 tests — **100%** intent classification, tool-call accuracy, hallucination-free — at **6.4s** avg latency.

TECHNICAL SKILLS

Languages: Python, C++, JavaScript/TypeScript, SQL

ML/DL: PyTorch, TensorFlow, Transformers, PEFT/LoRA, Scikit-learn, NumPy, Pandas

LLM & Agents: LangGraph, LangChain, RAG, Prompt Engineering, Tool-Calling, Generative AI

Retrieval: Qdrant, FAISS, Hybrid Search (BM25 + Dense), Reranking

Backend & Infra: FastAPI, Docker, PostgreSQL, Next.js, Linux, Git/GitHub, Render

MLOps & Observability: LangSmith, MLflow, Weights & Biases, GitHub Actions

Evaluation: LLM-as-Judge, ROUGE, BERTScore, METEOR, Ragas

Certifications: Deep Learning Specialization (Coursera), LangChain for LLM Application Development (deeplearning.ai)